Acids, Bases & Buffers Cheat Sheet: The 6 Ways to Determine pH

1) Strong Acids/ Strong Bases

You MUST memorize:

Strong Acids: HCl, HBr, HI, HNO_3, $HClO_4$, H_2SO_4

Strong Bases: Groups IA and IIA metal hydroxides

100% Dissociation! Easy life:

$$pH = -\log[H^+] = -\log[HA]_o$$

$$pOH = -\log[OH^-] = -\log[B]_o$$

$$pH + pOH = 14$$

2) Weak Acids/ Weak Bases

If it's not strong, it's weak!

< 1% Dissociation → Equilibrium!

Time saver!! Since acids ionize 1 H^+ at a time, $[H_3O^+] = [A^-]$, and $[OH^-] = [BH^+]$. For weak acids and bases, make the assumption $[HA]_0 - \boldsymbol{x} \approx [HA]_0$ and $[B]_0 - \boldsymbol{x} \approx [B]_0$.

Weak Acids:

$$K_a = \frac{[x][x]}{[HA]_o - x} \approx \frac{[x][x]}{[HA]_o} \text{ where } [H_3O^+] = x << [HA]_o$$

Weak Bases:

$$K_b = \frac{[x][x]}{[B]_o - x} \approx \frac{[x][x]}{[B]_o} \text{ where } [OH^-] = x << [B]_o$$

3) Salty Salts

To know if a salt will affect pH, determine: → Will the salt ions will hydrolyze (or split) water?

Conjugates of Strong Acids/Bases: do **NOT** hydrolyze water, and thus don't affect pH

Conjugates of Weak Acids/Bases: DO hydrolyze water, and thus do affect pH!

How to Determine the pH of a Salt	Example
1. Dissociate your salt.	$CaSO_3 \rightarrow Ca^{2+} + SO_3^{2-}$
Make the cation into a base: is it strong or weak?	$Ca^{2+} + 2\ OH^- \rightarrow Ca(OH)_2$ **Strong base**
2. Make the anion into an acid: is it strong or weak?	$H^+ + SO_3^- \rightarrow HSO_3^-$ **Weak acid**
3. **Strong wins!**	SB + WA, so this salt is basic!
4. If either is weak, write the hydrolysis reaction:	$SO_3^- + H_2O \rightleftharpoons OH^- + HSO_3^-$

Conjugate base of WA: $A^- + H_2O \rightleftharpoons OH^- + HA$

Conjugate acid of WB: $BH^+ + H_2O \rightleftharpoons H_3O^+ + B$

5. Use your hydrolysis equation to calculate the pH using the **Weak Acids/Bases** method.
6. ***Be careful***. Did the problem give you K_a, or K_b instead? Do you need to convert based on your hydrolysis reaction? Remember: $K_w = K_a \times K_b = 1.0 \times 10^{-14}$

4) Adding Strong Acids + Strong Bases (Yes, this includes titrations!)

These are really just stoichiometry problems with a limiting reagent! ☺

- **What is "excess" determines the pH** (since all strong/strong combos neutralize)
- **Beware the change in volume if multiple solutions are being added**! (calculate moles of either H^+ or OH^- in excess, and divide by total volume to determine concentration of $[H^+]$ or $[OH^-]$, and go from there)

5) Buffers

Whenever a weak acid or base is present with its conjugate salt – YOU HAVE A BUFFER!!! Four ways to get a buffer:

1. Weak acid and its conjugate base (HA and A^- **OR** HA and *Na*A) **1:1 mole ratio**
2. Weak base and its conjugate acid (B and BH^+ **OR** B and BH*Cl*) **1:1 mole ratio**
3. Weak base with strong acid (titration) **1 WB : 0.5 SA mole ratio**
   → strong acid reacts with weak base, producing conjugate acid
4. Weak acid with strong base (titration) **1 WA : 0.5 SB mole ratio**
   → strong base reacts with weak acid, producing conjugate base

The <u>best buffer</u> has: 1) High capacity (lots of acid and base), **2) [HA] = [A^-], 3) pH (of buffer) = pK_a (of acid form)**

Two calculation options: $[H_3O^+] = K_a \dfrac{[HA]}{[A^-]}$ **or** $\text{pH} = \text{pKa} + \log\left(\frac{[A^-]}{[HA]}\right)$

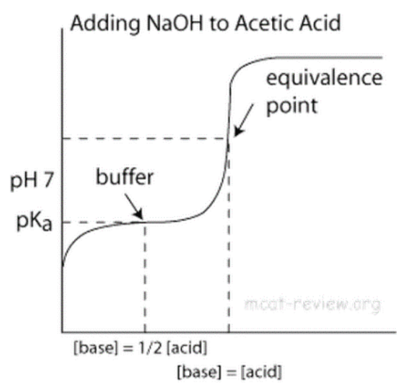
[HA] = Weak acid **or** salt of conjugate base **or** added strong base (in a WB/SA titration)
[A^-] = Weak base **or** salt of conjugate acid **or** added strong acid (in a WA/SB titration)

Shortcut!!! Since $\frac{[Acid]}{[Base]}$ is a ratio in the equations, the <u>amount of moles may be substituted in place of concentration</u> because the final volumes will be the same, and thus cancel out.

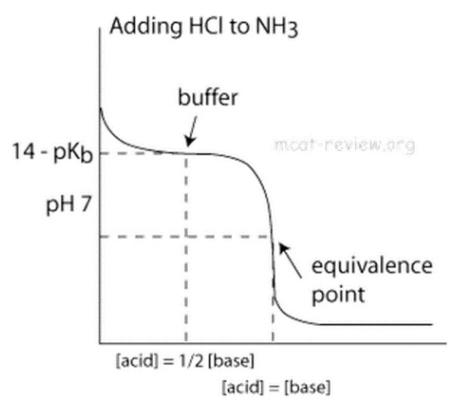
6) Adding Strong Acids/Bases + Weak Bases/Acids (Yes, this includes titrations!)

Buffers make the titration curve "flat" at the region where buffering occurs. On a titration curve, this is the point of inflection (buffer arrow) = maximum buffering capacity!

Weak Acid Titrated with Strong Base



Weak Base Titrated with Strong Acid



There are 4 zones of interest along a titration curve for a Weak Acid/Base and Strong Base/Acid:

1. <u>Initial pH</u>: simply a weak acid or weak base problem!
2. <u>During titration, before equivalence point</u>: BUFFER! The amount of conjugate formed = the amount of strong acid or base added. (Amount of weak left = HA_o − strong added)
3. <u>Equivalence point</u>: only salt and water are left; salt is either a weak base (if a weak acid was titrated) or a weak acid (if a weak base was titrated); remember to convert K_a to K_b or vice versa! (Use $M_AV_A = M_BV_B$)
4. <u>Beyond the equivalence point</u>: Stoichiometry! Calculate how much STRONG (acid or base) is left after equivalence, recalculate its molarity (remember, volume increased during titration), and calculate using the strong acid/base method.

<u>Best Indicator</u>: Choose indicator with pK_a (**of indicator**) ≈ pH (**at equivalence point of titration**). This means that

K_a of the indicator ≈ $1 \times 10^{-\text{pH @ eq pt}}$)

<u>Remember</u>: **if pH ≤ pK_a** the acid form (HA) predominates, **if pH > pK_a** the conjugate base form (A^-) predominates

Unit 7.1 Multiple Choice Practice

1. How many liters of distilled water must be added to 10. mL of an aqueous solution of HNO_3 with a pH of 2 to create a solution with a pH of 3?

   a. 10. mL b. 20. mL c. 40. mL d. 90. ml

2. The $[OH^-]$ in a solution with a pH of 3.00 is

   a. 1.0×10^{-11} M b. 1.0×10^{-9} M c. 1.0×10^{-6} M d. 1.0×10^{-3} M

3. Consider the following:

   I. H_2SO_4
   II. HSO_4^-
   III. SO_4^{2-}

   Which of the chemical species above are present in a reagent bottle labeled 1.0 M $H_2SO_4(aq)$?

   a. I only b. I and II only c. II and III only d. I, II, and III

4. The pH of a 1.0 mL solution of 0.30 M NaOH if 9.0 mL of distilled water is added to the solution is closest to:

   a. 3 b. 11 c. 12 d. 13

5. A 0.25 M solution of a weak monoprotic acid has a pH of 6. What is the value of K_a for the acid?

   a. 4×10^{-12} b. 4×10^{-11} c. 2.5×10^{-6} d. 2.5×10^{-5}

6.. Consider the following equilibrium: $2\ H_2O(l) + \text{energy} \leftrightarrow H_3O^+(aq) + OH^-(aq)$

Which of the following describes the result of decreasing the temperature?

	$[H_3O^+]$	$[OH^-]$	K_w
a.	increases	increases	increases
b.	decreases	increases	decreases
c.	increases	decreases	no change
d.	decreases	decreases	decreases

7. The pH of a solution changes from 3.00 to 6.00. By what factor does the $[H_3O^+]$ change?

a. 3 b. 30 c. 100 d. 1000

$$HClO_2(aq) + C_2H_3O_2^-(aq) \leftrightarrow ClO_2^-(aq) + HC_2H_3O_2(aq)$$

8. The equilibrium constant for the reaction represented by the equation above is greater than 1.0. Which of the following gives the correct relative strength of the acids and bases in the reaction?

	Acids	Bases
a.	$HC_2H_3O_2 > HClO_2$	$ClO_2^- > C_2H_3O_2^-$
b.	$HClO_2 > HC_2H_3O_2$	$C_2H_3O_2^- > ClO_2^-$
c.	$HClO_2 > HC_2H_3O_2$	$ClO_2^- > C_2H_3O_2^-$
d.	$HC_2H_3O_2 > HClO_2$	$C_2H_3O_2^- > ClO_2^-$

9. When 10.0 mL of 0.10 M HCl is added to 10.0 mL of water, the concentration of H_3O^+ in the final solution is:

a. 0.010 M b. 0.050 M c. 0.10 M d. 0.20 M

10. A 0.50 M solution of a weak monoprotic acid has a pH of 4. Calculate the ionization constant, K_a, for the acid.

a. 5×10^{-9} b. 2×10^{-8} c. 2×10^{-4} d. 5×10^{-5}

11. A bottle of water is placed in an ice bath to chill. What happens to the pH of the water as it cools?

a. It will increase because the auto-ionization of water is an endothermic process.

b. Nothing; the volume would have to change in order for any ion concentration to change.

c. Nothing; pure water always has a pH of 7.00.

d. It will decrease because the concentration of H^+ is decreasing.

12. In order to change the pH of a solution from 2.0 to 4.0 the $[H_3O^+]$ must

a. increase by a factor of 2

b. decrease by a factor of 2

c. increase by a factor of 100

d. decrease by a factor of 100

13. What is the conjugate acid and the conjugate base of HPO_4^{2-}?

	Conjugate Acid	Conjugate Base
a.	PO_4^{3-}	$H_2PO_4^-$
b.	$H_2PO_4^-$	PO_4^{3-}
c.	$H_2PO_4^-$	H_3PO_4
d.	H_3PO_4	PO_4^{3-}

14. When comparing 1.0 M solutions of bases, the base with the lowest $[OH^-]$ is the

a. weakest base and it has the largest K_b value.

b. strongest base and it has the largest K_b value.

c. weakest base and it has the smallest K_b value.

d. strongest base and it has the smallest K_b value.

15. What is the concentration of $Sr(OH)_2$ in a solution with a pH = 11.00?

a. 2.0×10^{-11} M   b. 1.0×10^{-11} M   c. 5.0×10^{-4} M   d. 1.0×10^{-3} M

For which of the reactions above is the **net ionic** equation $H^+ + OH^- \rightarrow H_2O$?

16. Consider the neutralization reactions between the following acid-base pairs in dilute aqueous solutions:

(1) $CH_3COOH + NaOH \rightarrow$

(2) $HNO_3 + KOH \rightarrow$

(3) $H_3PO_4 + Ba(OH)_2 \rightarrow$

(4) $HCl + KOH \rightarrow$

(5) $H_2SO_4 + NH_3 \rightarrow$

For which of the reactions above is the **net ionic** equation $H^+ + OH^- \rightarrow H_2O$?

a. 1, 3   b. 2, 4, 5   c. 2, 4   d. 3, 5

FR Practice #2 (2018 #5, 4 points)

$$HF(aq) + H_2O(l) \leftrightarrow F^-(aq) + H_3O^+(aq)$$

2. The ionization of HF(aq) in water is represented by the equation above. In a 0.0350 M HF(aq) solution, the percent ionization of HF is 13.0 percent.

   a. Two particulate representations of the ionization of HF molecules in the 0.0350 M HF(aq) solution are shown below in Figure 1 and Figure 2. Water molecules are not shown. Explain why the representation of ionization of HF molecules in water in Figure 1 is more accurate than the representation in Figure 2. (1 point)

   (The key below identifies the particles in the representations.)

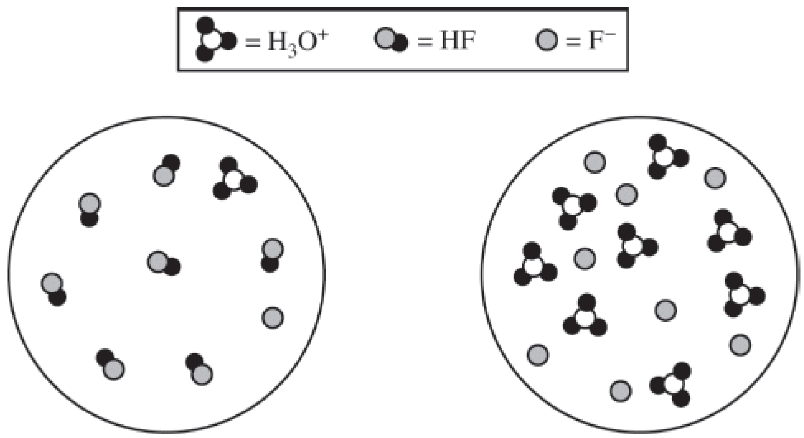


Figure 1 Figure 2

   b. Use the percent ionization data above to calculate the value of K_a for HF. (2 points)

   c. If 50.0 mL of distilled water is added to 50.0 mL of 0.0350 M HF(aq), will the percent ionization of HF(aq) in the solution increase, decrease, or remain the same? Justify your answer with an explanation or a calculation. (1 point)

FR Practice #3 (2016 #4, 4 points)

$$C_6H_5OH(aq) + H_2O(l) \rightleftarrows C_6H_5O^-(aq) + H_3O^+(aq) \qquad K_a = 1.12 \times 10^{-10}$$

3. Phenol is a weak acid that partially dissociates in water according to the equation above.

   a. What is the pH of a 0.75 M $C_6H_5OH(aq)$ solution? (2 points)

Free Response Practice #4 (5 points)

A pure 14.85 g sample of the weak base ethylamine, $C_2H_5NH_2$, is dissolved in enough distilled water to make 500. mL of solution.

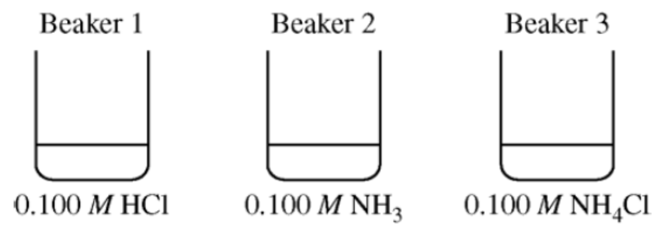
(a) Calculate the molar concentration of the $C_2H_5NH_2$ in the solution. (2 points)

(b) Write an equation showing how the aqueous ethylamine reacts with water. (1 point)

(c) Write the equilibrium-constant expression for the reaction between $C_2H_5NH_2(aq)$ and water. (1 point)

(d) Of $C_2H_5NH_2(aq)$ and $C_2H_5NH_3^+$ (aq), which is present in the solution at the higher concentration at equilibrium? Justify your answer. (1 point)

Free Response Practice #5 (2011 #1, 10 points)



2. Each of three beakers contains 25.0 mL of a 0.100 M solution of HCl, NH_3, or NH_4Cl, as shown above. Each solution is at 25°C.

   a. Determine the pH of the solution in beaker 1. Justify your answer. (1 point)

   b. In beaker 2, the reaction $NH_3(aq) + H_2O(l) \leftrightarrow NH_4^+(aq) + OH^-(aq)$ occurs. The value of K_b for $NH_3(aq)$ is 1.8×10^{-5} at 25°C.

      i. Write the K_b expression for the reaction of $NH_3(aq)$ with $H_2O(l)$. (1 point)

      ii. Calculate the $[OH^-]$ in the solution in beaker 2. (2 points)

c. In beaker 3, the reaction $NH_4^+(aq) + H_2O(l) \leftrightarrow NH_3(aq) + H_3O^+(aq)$ occurs.

i. Calculate the value of K_a for $NH_4^+(aq)$ at 25°C. (1 point)

ii. The contents of beaker 2 are poured into beaker 3 and the resulting solution is stirred. Assume that volumes are additive. Calculate the pH of the resulting solution. (2 points)

d. The contents of beaker 1 are poured into the solution made in part (c)(ii). The resulting solution is stirred. Assume that volumes are additive. Is the resulting solution an effective buffer? Justify your answer. (1 point)